

EUROAVIA
AIRBUS SLOSHING ROCKET WORKSHOP 2026

FINAL DESIGN REPORT

Project Buzz Lightyear
“In Slosh We T(h)rust”



Team Members

Brian Bertacco	Sloshing CFD Lead
Tobia De Benedictis	Aerodynamics & Stability
Anna Testani	Flight Dynamics & Propulsion
Lucia Tola	Flight Dynamics & Propulsion
Felice Venuti	Aerodynamics & Stability

Submission Date: 14 June 2026

Word Count (Introduction–Conclusion): [XXXX]

Abstract

Abstract

This report presents the Final Design of Project Buzz Lightyear, developed by team In Slosh We T(h)rust for the ASRW 2026 competition. Building on a preliminary study that scored 85/100, the report refines and validates the deployable-wing glider water rocket concept (Configuration C) selected to maximise the competition Flight Score $F_S = (\sqrt{HD^2 + Alt^2} + T) \times m_{payload} / m_{TOW}$. The vehicle employs dual 1.5 L PET propulsion bottles and a dedicated sloshing tank filled to 50 % capacity (750 mL). Baffling is achieved through a Mixed configuration (annular rings with a central cruciform mast, 3D-printed in PLA, transverse blockage 48.9 %), chosen for targeted lateral-mode suppression and confirmed REQ-11 compliance. External aerodynamic CFD in ANSYS Fluent yielded for the folded wings configuration $C_{D0} = 0.01$ and a lift-to-drag ratio of 2.73 at 10° , while for the extended wings configuration the following values were computed $C_{D0} = 0.016$ and a lift-to-drag ratio of 4.44 at 10° . MATLAB trajectory simulations predict a maximum altitude of 38.94 m, a horizontal range of 18.28 m, and a Flight Score of 18.41. Sloshing CFD analyses using the Volume-of-Fluid (VOF) method established a baseline free-surface oscillation frequency of 3.15 Hz in the unbaffled configuration; baffled simulation results demonstrate a 72.7 % reduction in peak lateral sloshing force. A full Design Verification and Validation assessment demonstrates compliance with all 17 mandatory requirements, and a risk assessment conducted with the ASRW template confirms operational safety for both manufacturing and launch operations.

Keywords: water rocket, sloshing, VOF CFD, deployable-wing, baffle design, flight score optimisation.

Contents

Abstract	1
List of Tables	4
List of Figures	5
Nomenclature	6
1 Introduction	1
1.1 Background and Literature Review	1
1.2 Scope and Objectives	2
1.3 Project Organisation	2
1.3.1 Team Structure	2
1.3.2 Project Timeline	3
1.3.3 Progress Since the Conceptual Design Report	3
2 Requirements Capture	4
2.1 Design Requirements	4
2.2 Assumptions	4
2.3 Success Criteria	4
3 Concept Design	5
3.1 Design Exploration	5
3.1.1 Configuration A — Classic Rocket with Parachute Recovery (Discarded)	6
3.1.2 Configuration B — Delta-Wing Glider (Discarded)	6
3.1.3 Configuration C — Deployable Wings and Secondary Surfaces (Selected)	6
3.2 Configuration Selection	6
4 Detailed Design	7
4.1 General Arrangement	7
4.2 Aerodynamic Design	7
4.2.1 Wing Geometry	7
4.2.2 Stability Margins	8
4.2.3 External Aerodynamics CFD	8
4.3 Propulsion System	10
4.3.1 System Overview	10
4.3.2 Pressure Rating of the Internal Plumbing	11
4.3.3 Rectus Type 26 Quick Coupler Integration	11
4.4 Sloshing Tank Design	11
4.4.1 Tank Geometry	11
4.4.2 Baffle Design — Annular and Mixed Configurations	12
4.4.3 Baffle 3D Printing and Assembly Guide	13
4.5 Sloshing CFD Analysis	14
4.5.1 Methodology	14
4.5.2 Gravity Body-Force Profile	15
4.5.3 Mesh Statistics	15
4.5.4 Unbaffled Baseline Results	17
4.5.5 Baffled Results: Annular vs. Mixed Configuration	17
4.6 Flight Dynamics	19
4.6.1 Trajectory Model	19

4.6.2	Propulsion Data	20
4.6.3	Trajectory Simulation Results	20
4.7	Wing Deployment Mechanism	21
4.8	Altimeter Casing	21
4.9	Manufacturing Plan	22
4.9.1	Bill of Materials	22
4.9.2	Component Manufacture	24
4.9.3	Assembly Sequence	24
4.10	Safety Assessment	24
4.10.1	Manufacturing and Assembly Risks	25
5	Design Verification and Validation	25
5.1	External Aerodynamics CFD Validation	25
5.1.1	Grid Convergence Study	25
5.1.2	Analytical Cross-Validation	26
5.2	Sloshing Validation	27
5.2.1	Grid Convergence Study	27
5.2.2	Sloshing Frequency Validation	28
5.3	Test Plan	29
5.4	Design Compliance Matrix	30
6	Conclusion	30
	References	33
A	Full Risk Assessment Matrix	34
B	Trajectory Simulation Input Parameters	34
C	Sloshing CFD Named Expressions	34

List of Tables

1	Team structure and FDR deliverable ownership.	3
2	Full requirements capture and FDR compliance status.	5
3	Concept trade-off summary.	7
4	Finalised vehicle dimensions.	8
5	Configuration C wing and fin geometry.	8
6	Configuration C — CoG and CoP positions (measured from nose tip, positive aft).	8
7	External aerodynamics CFD mesh statistics.	9
8	External aerodynamics CFD simulation results for Configuration A.	10
9	External aerodynamics CFD simulation results for Configuration B.	10
10	Sloshing tank geometric parameters.	11
11	Baffle geometry and REQ-11 compliance verification.	12
12	ANSYS Fluent solver configuration for sloshing simulations.	15
13	Sloshing CFD mesh statistics for unbaffled and baffled configurations.	16
14	Unbaffled tank CFD results.	17
15	Comparative sloshing CFD results: unbaffled vs. Annular vs. Mixed configurations.	17
16	Flight simulation results for Configurations A and B.	20
17	Bill of Materials (all prices in EUR, incl. VAT and shipping).	22
18	Manufacturing and assembly risk summary (excerpt from full risk matrix).	25
19	Launch and flight operations risk summary (excerpt from full risk matrix).	26
20	Grid convergence study: external aerodynamics CFD (Configuration B, C_D at $\alpha = 0^\circ$). Cell counts refer to the half-domain (symmetry plane applied, cf. Section 4.2.3).	26
21	Analytical cross-validation: C_D at $\alpha = 0^\circ$, Configuration A (body + three fins, $S_{\text{ref}} = 0.16\text{m}^2$).	27
22	Grid convergence study — sloshing CFD (unbaffled lateral force).	28
23	Sloshing frequency validation.	28
24	Test plan — verification method and acceptance criteria per requirement.	29
25	Design Compliance Matrix.	30
26	Final vehicle data sheet summary.	32
27	Full input parameter table for the water rocket trajectory simulation.	34

List of Figures

1	Project timeline (Gantt chart)	3
2	Three-view general arrangement drawing of the finalised Configuration C vehicle geometry. Dimensions: L = 951 mm, D = 94.4 mm, Wingspan = 556 mm.	7
3	Dimensioned cross-section at the point of maximum baffle blockage, demonstrating compliance with REQ-11 (Area $\leq 50\%$).	12
4	3D structural views of the implemented baffle designs. Top: Annular configuration. Bottom: Mixed (Annular + Cruciform) configuration.	14
5	Cross-sectional mesh topologies for the evaluated baffle configurations, highlighting poly-hexcore refinement.	16
6	Water volume fraction contours ($\alpha_w = 1$ in red, $\alpha_w = 0$ in blue) comparing the Annular and Mixed configurations.	18
7	Velocity magnitude distribution inside the tank, illustrating the superior kinetic energy dissipation of the Mixed configuration.	19
8	Cross-sectional view of the PLA altimeter casing, detailing the bolted flange, Teflon-sealed threads, mechanical O-ring seal, internal foam padding, and static pressure vent alignment.	22

Nomenclature

Symbol	Description	Unit
Alt	Maximum altitude achieved	m
C_D	Drag coefficient	–
C_L	Lift coefficient	–
D	Body outer diameter	mm
F_D	Aerodynamic drag force	N
F_L	Aerodynamic lift force	N
f_1	First-mode lateral sloshing frequency	Hz
F_S	Competition flight score	–
g	Standard gravitational acceleration	m/s ²
h	Fill depth in sloshing tank	m
h_b	Baffle fin height	mm
HD	Total horizontal distance	m
L_b	Baffle fin length	mm
m_{payload}	Payload mass	kg
m_{TOW}	Takeoff gross mass	kg
P_0	Initial gauge pressure	bar
R	Inner radius of sloshing tank	m
Re	Reynolds number	–
T	Total flight time	s
t_b	Baffle fin thickness	mm
TWR	Thrust-to-weight ratio	–

Abbreviations

AoA	Angle of Attack
CDR	Conceptual Design Report
CFD	Computational Fluid Dynamics
CoG	Centre of Gravity
CoP	Centre of Pressure
FDR	Final Design Report
FFT	Fast Fourier Transform
FMV	Fair Market Value
PET	Polyethylene Terephthalate
PETG	Polyethylene Terephthalate Glycol
VOF	Volume of Fluid
WRS	Water Rocket Simulator

1 Introduction

1.1 Background and Literature Review

Introduction to Liquid Sloshing in Aerospace

Liquid sloshing is the dynamic motion of a free surface inside a partially filled container subject to external excitation. This challenge is encountered across the full spectrum of aerospace vehicles. From cryogenic upper stages of orbital launch vehicles to hypersonic atmospheric entry capsules, uncontrolled propellant oscillations can couple with structural flexibility modes, degrade attitude control authority and in extreme cases drive vehicle instability. The problem was first systematically cataloged by Abramson (1966) in NASA SP-106[1]. Recently, renewed interest in the field has been driven by the rapid growth of small-satellite launch vehicles, whose slender tanks, that are optimised for mass fraction, operate at fill levels that inadvertently maximise sloshing coupling risk.

Theoretical foundations

Classical sloshing theory relies on the assumption of irrotational, incompressible and inviscid fluid [7]. The fundamental natural frequency for the first asymmetric sloshing mode in a rigid circular cylindrical tank of inner radius R with fill depth h is given by:

$$f_1 = \frac{1}{2\pi} \sqrt{\frac{1.841 g}{R} \tanh\left(1.841 \frac{h}{R}\right)} \quad (1)$$

where 1.841 is the first root of the derivative of the first-order Bessel function of the first kind [8]. At the 50 % fill level mandated by the competition regulations (where $h = 158.5$ mm and $R = 47.2$ mm, placing the fluid in the deep-tank regime where $\tanh \approx 1$), this evaluates to $f_1 \approx 3.11$ Hz for the project tank (eq. (19)), a value subsequently validated with CFD results.

Slosh Mitigation and Baffle design

Passive baffling remains the most mechanically robust approach to sloshing suppression, avoiding the complexity of active control systems. Ring baffles, recommended in NASA SP-106, primarily target axial modes and are less effective against transverse oscillations. Longitudinal cruciform fins, conversely, directly partition the convective liquid mass to attenuate dominant lateral modes. By integrating both into a **Mixed** hybrid architecture, the design exploits annular rings to disrupt axial boundary layer flows while utilising a central cruciform mast to suppress the lateral sloshing that couples with pitch and yaw dynamics during rocket ascent [1].

To predict these highly non-linear fluid dynamics, modern engineering relies heavily on numerical methods. High-fidelity Volume-of-Fluid (VOF) simulations with ANSYS Fluent have been validated against experimental sloshing data in multiple recent studies, confirming their suitability for free-surface flow with large interface deformations [2].

Design inspiration

Beyond the aerospace domain, the biomechanics field offers instructive analogies: the role of trabecular bone microstructure in damping fluid oscillations within long bones shares topological similarity with the multi-axis fluid partitioning of the proposed mixed baffle design. Similarly, the internal baffling of industrial fuel tankers on road vehicles (regulated under ADR standards) demonstrates proven effectiveness of longitudinal dividers in suppressing lateral slosh under lateral acceleration profiles comparable to those experienced during rocket ascent.

1.2 Scope and Objectives

The primary scope of Project Buzz Lightyear is the design, simulation, and validation of a pressurized water rocket optimized to maximize the Flight Score parameter. The core project objectives are to maximize horizontal range and altitude utilizing a lifting aerodynamic architecture, while mitigating destabilizing lateral sloshing modes.

To demonstrate the achievement of these project goals, the objectives of this Final Design Report are to:

- (i) Present the finalised aerodynamic configuration (Configuration C, deployable-wing glider) with updated simulation progress regarding geometry, stability margins, and overall CFD validation;
- (ii) Document the sloshing analysis performed by the team, establishing the unbaﬄed baseline. This includes a comparison between purely Annular baffles and the Mixed hybrid configuration via CFD results, evaluated against the baseline to finalise the design choice;
- (iii) Provide geometric, analytical, and numerical evidence of compliance with the requirements;
- (iv) Present the full manufacturing plan, Bill of Materials, and risk assessment;
- (v) Report the Design Compliance Matrix as the final step of the design validation process, alongside the Final Data Sheet summarizing the chosen baseline.

The results reported in this document were generated through simulations using ANSYS Fluent (sloshing VOF and external aerodynamics CFD) and MATLAB/Simulink (6-DoF trajectory simulation).

1.3 Project Organisation

To ensure a manageable workload for all team members, all concurrently completing the final stages of their master's degrees, the project was structured around a balanced timeline. A core element of this organization was the strategic division of roles. While all members share a comprehensive aerospace engineering background, specific tasks were allocated based on individual expertise and academic interests. Tasks were divided accordingly, leveraging the specialized knowledge of members focused on aerodynamics for CFD and trajectory, and members focused on systems engineering and sloshing dynamics. To mitigate potential delays caused by overlapping academic commitments or any possible inconvenience related to simulations, the timeline incorporated contingency buffers, avoiding unfeasible rush periods.

1.3.1 Team Structure

As previously outlined, team members were assigned to the different roles based on their individual academic backgrounds and specialized interests. Table 1 summarizes the core responsibilities of each member and their primary deliverable ownership tailored for the Final Design Report phase.

Table 1: Team structure and FDR deliverable ownership.

Member	Role	Primary Discipline	Activities
Tobia De Benedic- tis, Felice Venuti	Aerodynamics	Aerodynamics & Stability	Final wing geometry V&V, external CFD validation, altimeter mounting integration
Anna Testani, Lucia Tola	Flight Dynamics	Dynamics Simulation & Propulsion	Final MATLAB trajectory simulation, BoM, Risk Analysis, Final Data Sheet
Brian Bertacco	Sloshing Analysis	CFD Simulation & Fluid Dynamics	Baffled tank CFD analysis (Design 1 & 2), structural blockage proofs, risk assessment

1.3.2 Project Timeline

Figure 1 presents the project Gantt chart covering the period from the CDR submission to the Final Design Report deadline.

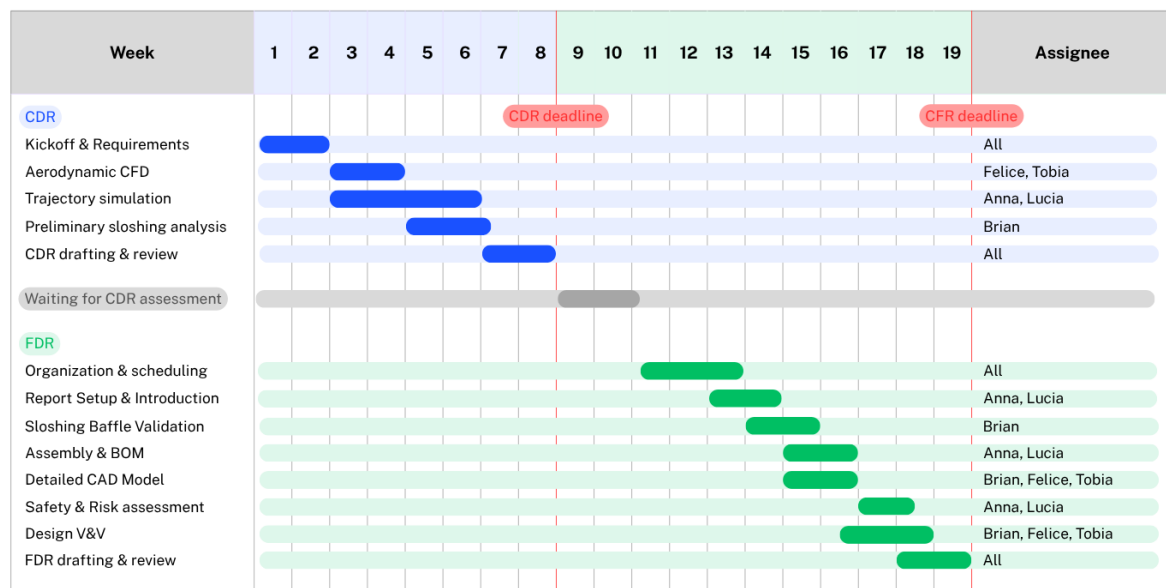


Figure 1: Project timeline (Gantt chart)

Contingency buffers of two weeks are built into the CFD simulation campaign and the FDR drafting phase to absorb potential delays from mesh convergence issues or simulation reruns following boundary condition updates.

1.3.3 Progress Since the Conceptual Design Report

The Conceptual Design Report (CDR) was awarded a score of 85/100. The key development steps undertaken since the submission include:

- Baffle design finalisation: the mixed configuration, which integrates annular rings with a central

cruciform mast, was confirmed as the primary internal architecture. To eliminate the difficulty of inserting components through the restrictive 28 mm bottle neck, the assembly methodology was updated to rely on opening the base of the tank, allowing the baffle to be 3D-printed as a single unified PLA structure and inserted as one complete module.

- Baffled CFD campaign: ANSYS Fluent Volume-of-Fluid (VOF) simulations evaluating the baffled tank were completed. The mixed design demonstrated superior liquid isolation and kinetic energy dissipation.
- Flight dynamics simulation: The MATLAB trajectory simulation was advanced to model three distinct phases: powered flight, ballistic coast, and glide descent. The updated model accounts for the aerodynamic contributions of the canards and tail surfaces to accurately predict the vehicle's final flight score.
- Manufacturing plan: a comprehensive assembly sequence, a complete Bill of Materials, and a cost breakdown were finalized to ensure the physical test article can be manufactured within the established budget.
- Safety assessment: a full Risk Assessment was completed using the provided ASRW template.

2 Requirements Capture

2.1 Design Requirements

Table 2 presents the complete requirements capture derived from the ASRW 2026 Technical Regulations, including the team's defined success criterion and compliance status at FDR stage.

2.2 Assumptions

The following assumptions underpin the design and analysis:

- The PET bottle geometry is approximated as a perfect circular cylinder of inner radius $R = 47.2$ mm and height 317 mm for analytical sloshing frequency calculation. The actual tapering at the shoulders introduces a conservative overestimate of sloshing frequency.
- Water is treated as incompressible and isothermal at 20 °C throughout all CFD simulations.
- The launch rail imparts no lateral perturbation; the 85° initial attitude is the sole excitation source for the sloshing analysis.
- Structural analysis assumes the main body and propulsion bottles are manufactured from commercially available 1.5 L PET bottles ($t_{\text{wall}} \approx 0.3$ mm).
- The payload is represented as a 30 g point mass positioned at the nose.

2.3 Success Criteria

The team defines mission success as:

- (i) A Flight Score $F_S > 15$ under competition conditions;
- (ii) Full compliance with REQ_01–REQ_17;
- (iii) Maximum lateral sloshing force in the baffled configuration $< 50\%$ of the unbaffled baseline.

Table 2: Full requirements capture and FDR compliance status.

ID	Shall Statement	Verification Method	Status
REQ_01	Total vehicle length ≤ 1.5 m	Physical measurement	PASS
REQ_02	Wingspan ≤ 2.25 m	Physical measurement	PASS
REQ_03	Takeoff mass ≤ 5 kg	Scale reading	PASS
REQ_04	Sloshing tank ≥ 500 mL at 50 % fill; fill mark visible	Volume measurement; visual	PASS
REQ_05	Propellant mass $\leq$ sloshing liquid mass	Mass measurement	PASS
REQ_06	Launch angle $85 \pm 3^\circ$	Inclinometer	TBC
REQ_07	Pressurisation ≤ 10 atm	Pressure gauge	PASS
REQ_08	Rectus Type 26 Quick Coupler male fitting	Visual inspection; coupling test	PASS
REQ_09	Air as inflation gas only; water at ambient temperature as propellant	Inspection of inflation source	PASS
REQ_10	Remote launch from ≥ 5 m	Umbilical length measurement	TBC
REQ_11	Baffle cross-section ≤ 50 % of tank inner area (per axis)	CAD + CFD verification	PASS
REQ_12	Altimeter: water-resistant; 3×3 mm ² vents; no adhesive	Visual inspection by judges	PASS
REQ_13	Total budget $\leq$ €500 (incl. 15 % buffer); no single COTS item $>$ €300	Bill of Materials	PASS
REQ_14	Tanks presented to judges for volume verification before integration	Pre-integration inspection	TBC
REQ_15	Maximise flight score (Eq. 1)	Measured flight data	TBC
REQ_16	Reusable for ≥ 3 launches	Post-flight structural check	TBC
REQ_17	Sloshing tank min. cross-section ≥ 50 % of max cross-section (per axis)	CAD measurement	PASS

3 Concept Design

3.1 Design Exploration

The architectural exploration was driven by the structure of the competition Flight Score, which rewards horizontal range alongside altitude and flight time. Since range is maximised by a controlled gliding descent at a high lift-to-drag ratio, the design problem reduces to achieving a stable glide passively, without active control. For a statically stable vehicle ($dC_m/d\alpha < 0$), a trimmed glide at a useful positive angle of attack ($\alpha_{trim} > 0$) is possible only if the configuration generates a nose-up pitching moment at zero incidence, $C_{m,0} > 0$. This requirement is the central criterion against which the three candidate architectures were judged: a ballistic body recovered by parachute (Configuration A), a fixed lifting wing (Configuration B), and deployable wings augmented by secondary surfaces (Configuration C).

3.1.1 Configuration A — Classic Rocket with Parachute Recovery (Discarded)

Configuration A adopts a conventional water rocket geometry: a single-axis slender body with three stabilising fins and a nose-ejected parachute for recovery. The aerodynamic drag coefficient was established at $C_D = 0.01$ via CFD. The parachute substantially increases the effective drag area during descent, prolonging flight time but also limiting horizontal range. The predicted flight score of 14.77 places this configuration below the team’s success criterion.

Key parameters: $D = 94.4$ mm, $L = 951$ mm, reference area $S = 6998.96$ mm². Stability margin is 323 mm (measured from the nozzle), positive and sufficient for passive stability.

3.1.2 Configuration B — Delta-Wing Glider (Discarded)

Configuration B introduces two fixed delta main wings (span 494 mm, root chord 450 mm, area 0.11 m²) and three axisymmetric stabilising fins (span 294 mm, root chord 200 mm). While CFD results ($C_D = 0.016$ at 0° AoA; $L/D = 4.91$ at 5°) and trajectory simulations predicted a competitive flight score of 16.82, the fixed geometry incurs a severe drag penalty during the high-speed boost phase. Furthermore, the necessity of achieving a non-zero trim angle post-burnout required additional complex control elements. Consequently, this fixed-wing approach was discarded.

3.1.3 Configuration C — Deployable Wings and Secondary Surfaces (Selected)

To eliminate the drag penalty of fixed lifting surfaces during the powered ascent, this configuration employs deployable main wings alongside extendable secondary surfaces. The surfaces remain fully stowed during the boost phase to minimise drag, and deploy at an iteratively chosen point in the trajectory to enable a high-efficiency, extended glide phase.

While the mechanical complexity and the dimensional constraint of passing all components through the 28 mm bottle neck initially posed significant risks, the substantial aerodynamic advantages and potential to maximise F_S drove the selection of this architecture. The conceptualisation and mechanical design of both the primary and secondary extendable surfaces underwent several critical developmental iterations to ensure structural integrity and reliable deployment, which are detailed in Section 4.7.

3.2 Configuration Selection

Configuration A, lacking any lifting surface, cannot generate the gliding range the Flight Score rewards and falls below the success-criterion score; it was therefore eliminated at the architecture level, leaving Configurations B and C. The decisive difference between them is how, and whether, the positive zero-incidence moment identified in Section 3.1 can be produced passively. A fixed wing (Configuration B) is trimmed at 0° angle of attack: although its idealised CFD and trajectory results are competitive, sustaining a glide would require forcing a non-zero trim angle through additional control elements, in conflict with the passive-stability objective. Deploying the wing only after burnout would remove the boost-phase drag penalty, but deployment on its own does not alter the trim angle. Configuration C resolves this by adding secondary surfaces which, set at an appropriate incidence, generate the equivalent of a positive $C_{m,0}$ for the complete vehicle, establishing a stable trimmed glide passively, at the cost of greater mechanical complexity and the need to deploy at a specific point in the trajectory. The means of producing the required $C_{m,0} > 0$ was itself a design choice. A self-stabilising reflexed-camber wing provides it intrinsically but is prohibitively difficult to manufacture accurately at PET-bottle scale; a paraglider/parafoil naturally flies at positive incidence and is simple to build, but its flexible, non-rigid nature precludes reliable CFD characterisation; a rigid wing with one or more secondary surfaces supplies the moment through surface incidence, giving a tunable, simulable and intuitive solution at the cost of additional failure points and careful geometric tuning. The third option was adopted, directly defining Configuration C. The resulting trade-off between the two viable architectures, B and C, is summarised in Table 3.

Table 3: Concept trade-off summary.

Criterion	Config. B	Config. C (Selected)
Predicted Flight Score	16.82	18.41
Max Altitude (m)	37.69	38.94
Horizontal Range (m)	9.97	18.28
C_D (ascent)	0.016	0.010
L/D (glide)	1.72	1.72
Stability margin at launch (mm)	198	173
Recovery mechanism reliability	Passive (wings)	Passive (wings+secondary winglets)
Mechanical complexity	Low–Moderate	Moderate–High

4 Detailed Design

4.1 General Arrangement

The finalised vehicle geometry is presented in fig. 2. The rocket consists of three primary structural modules: (1) the nose cone and payload compartment, (2) the main fuselage housing the sloshing tank and avionics bay, and (3) the propulsion section comprising two interconnected 1.5 L PET bottles.

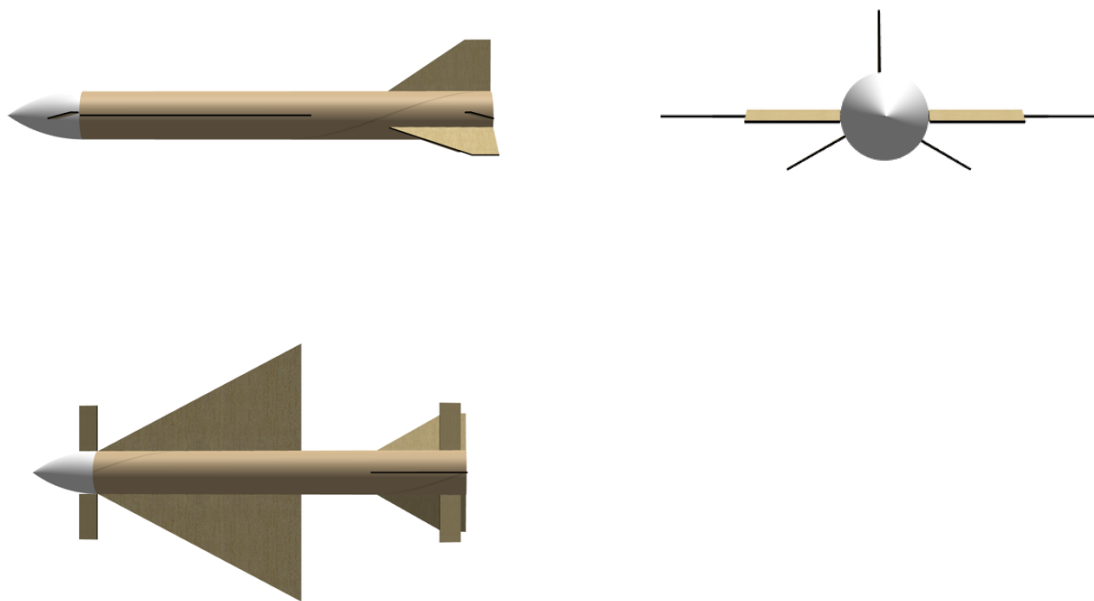


Figure 2: Three-view general arrangement drawing of the finalised Configuration C vehicle geometry. Dimensions: $L = 951$ mm, $D = 94.4$ mm, Wingspan = 556 mm.

Overall vehicle dimensions at the FDR stage are summarised in table 4.

4.2 Aerodynamic Design

4.2.1 Wing Geometry

The delta wing planform was selected for its low pitching moment coefficient and predictable stall behaviour at low Reynolds numbers. The wing incidence angle is set at 0° for the current design. A full

Table 4: Finalised vehicle dimensions.

Parameter	Value	Notes
Total length	951 mm	Nose tip to nozzle exit
Body outer diameter	94.4 mm	PET bottle $\varnothing$
Main wing full span	556 mm	Both panels combined
Stabilising fin span	294 mm	Per fin
Total lifting surface area	0.1975 m ²	Wings + fins
Sloshing tank height	317 mm	External

aerodynamic design summary is provided in Table 5.

Table 5: Configuration C wing and fin geometry.

Surface	Count	Span (m)	Root chord (m)	Area (m ²)	Pitch Angle (°)
Main wing (delta)	2	0.556	0.450	0.16	0
Stabilizing fins	3	0.294	0.200	0.0375	0
Canards	2	0.294	0.045	0.0132	20
Tail fins	2	0.294	0.045	0.0132	-25
Total	9	—	—	0.2239	—

4.2.2 Stability Margins

Longitudinal static stability requires the centre of gravity (CoG) to lie ahead of the centre of pressure (CoP). Table 6 tracks these positions throughout the flight.

Table 6: Configuration C — CoG and CoP positions (measured from nose tip, positive aft).

Flight Phase	CoG (mm)	CoP (mm)	Margin (mm)
At launch (full propellant)	508	681	+173
At burnout	362	681	+319
At wings release	362	430	+68
Mid-glide	362	430	+68

The CoP position varies due to the different aerodynamic configurations. The minimum positive margin of +68 mm during glide to ensure maneuverability, i.e. the effectiveness of the moment induced by the secondary surfaces.

4.2.3 External Aerodynamics CFD

Numerical simulations were performed to retrieve the aerodynamic performance of the proposed configurations in terms of C_D , C_L , and C_M across a range of angles of attack α . Only Configurations A (simple rocket, no wings) and B (rocket with delta wings) were simulated in CFD.

The aerodynamic contributions of the secondary surfaces (canards and tail) were not part of this CFD campaign. Instead, their effects were introduced directly through the MATLAB trajectory model (detailed in Section 4.6) following a "plug-and-play" philosophy. In this approach, the aerodynamic contribution of each individual secondary appendage is evaluated separately and then combined with the main body CFD data according to the principle of superposition.

Rather than re-meshing and re-running a full CFD campaign for every candidate geometry, this strategy allows the effect of any newly introduced appendage to be assessed rapidly, making explicit how each surface influences the complete configuration's performance. The method assumes negligible aerodynamic interference between surfaces; although this is an approximation—most notably regarding the downwash induced by forward surfaces on rearward ones—it was considered acceptable at this design stage, given the low angles of attack and the large streamwise separation between appendages.

Mesh quality. Both configurations were simulated in ANSYS Fluent through a parametric analysis in which the angle of attack was set as the input parameter. Each model was enclosed in a rectangular fluid domain sized so that the boundaries are sufficiently far from the body to capture the flow features developing around it, in particular the downstream wake, without the domain edges artificially constraining the solution or introducing blockage effects. To this end, the enclosure is asymmetric, with the largest extent allocated downstream of the model to fully accommodate the wake. Since both the geometry and the angle-of-attack sweep are symmetric about the vehicle's longitudinal vertical plane, a symmetry boundary condition was imposed on that plane so that only half of the domain had to be modelled, halving the cell count and substantially reducing the computational cost. Local mesh refinement was applied in the region of the model to increase accuracy where the flow gradients are strongest, balancing the trade-off between element count, and therefore computational cost, and solution accuracy. The refinement was carried out by progressively reducing the local element size on the body; since decreasing the element size drives a steep (near-exponential) growth in the total cell count, an iterative sizing study was performed until a converged final surface element size of 6×10^{-3} m was reached. Mesh quality was further assessed in terms of orthogonal quality, skewness and aspect ratio, reported in Table 7 for both configurations, and verified to lie within the recommended ranges as described by the ANSYS Guide [9]. Turbulence closure is provided by the $k-\omega$ Shear Stress Transport (SST) model. Lift and drag force monitors were tracked throughout each run; a solution was accepted when monitor variation fell below 0.1 % over the final 200 iterations and all scaled residuals were below 10^{-4} .

Table 7: External aerodynamics CFD mesh statistics.

Parameter	Config. A	Config. B
Aspect Ratio	1.85	1,850
Skewness	0.2312	0,2310
Orthogonality	0.7673	0,7675

Boundary conditions. The freestream inlet velocity is $v = 12.5$ m/s (negative z -direction), corresponding to the peak ascent velocity predicted by the trajectory simulation. This yields $Re = \rho v L / \mu = 192600$ (air at 15°C and $L = c_{ref}$ defined below). Outlet boundary conditions are pressure-far-field at zero gauge pressure.

Results. The results are summarised in Tables 8 and 9, for Configurations A and B, respectively. All coefficients were obtained in ANSYS Fluent using a common set of reference quantities: a reference area of $S_{ref} = 0.16$ m² and a reference chord of $c_{ref} = 225$ mm. The reference chord corresponds to the mean geometric chord of the tapered main wing. Adopting identical reference values for both configurations ensures that the resulting aerodynamic coefficients are directly comparable.

Table 8: External aerodynamics CFD simulation results for Configuration A.

α (°)	C_L	C_D	L/D
0	0	0.0101	0
10	0.0541	0.0198	2.73
20	0.0983	0.0456	2.15
30	0.119	0.0784	1.51
35	0.127	0.0986	1.28
40	0.125	0.116	1.08
50	0.116	0.144	0.80
70	0.0764	0.194	0.39
90	0.00553	0.206	0.03

Table 9: External aerodynamics CFD simulation results for Configuration B.

α (°)	C_L	C_D	L/D
0	0.00183	0.0160	0.11
5	0.128	0.0261	4.91
10	0.272	0.0612	4.44
20	0.519	0.198	2.62
25	0.627	0.299	2.10
30	0.701	0.408	1.72
35	0.704	0.496	1.42
40	0.640	0.538	1.19
45	0.579	0.576	1.00
50	0.526	0.622	0.85
55	0.465	0.658	0.71
60	0.405	0.695	0.58
65	0.345	0.728	0.47
70	0.280	0.753	0.37
75	0.209	0.761	0.27
80	0.145	0.778	0.19
85	0.0738	0.776	0.10
90	0.00693	0.777	0.01

4.3 Propulsion System

4.3.1 System Overview

The propulsion system comprises two 1.5 L PET bottles interconnected to form a unified propellant reservoir. The total internal volume is approximately 2.25 L, loaded with 0.75 L of water and pressurised with compressed air to an initial gauge pressure of 10 bar ($P_0 = 10\text{ bar}$). The two phases of propulsion are:

- **Water phase** ($t < 0.035\text{ s}$): high-thrust momentum flux from liquid water expulsion. Peak thrust approximately 1511 N; thrust-to-weight ratio $TWR = 77.2$.
- **Air phase** ($0.035 < t < 0.125\text{ s}$): secondary lower-magnitude thrust from pressurised air expansion.

4.3.2 Pressure Rating of the Internal Plumbing

The maximum working pressure is 10 bar gauge (11 bar absolute). The pressure rating of the 1.5 L PET bottle wall is verified using the thin-walled pressure vessel formula:

$$\sigma_{\text{hoop}} = \frac{P_{\text{max}} \cdot R_i}{t_{\text{wall}}} = \frac{1.1 \times 10^6 \text{ Pa} \times 0.0472 \text{ m}}{0.3 \times 10^{-3} \text{ m}} = 173 \text{ MPa} \quad (2)$$

The tensile strength of biaxially oriented PET is approximately 172–230 MPa [5]. At the design burst pressure (competition limit 10 bar), the theoretical hoop stress is at the lower bound of the yield range. Standard 1.5 L PET bottles are commercially rated and routinely tested to internal burst pressures exceeding 20 bar, providing a safety factor of > 2 at 10 bar operating pressure. This theoretical margin was empirically validated by the team through hydrostatic pressure testing of the assembled propulsion module up to 12 bar (120 % of the maximum operating pressure), which resulted in no visible yielding or pressure loss.

The interconnecting plumbing between the two bottles is constructed from custom 3D-printed PLA threaded adapters. These adapters are sealed with 30 mm nitrile O-rings and structurally reinforced at the bottle necks using marine epoxy. All joints are rated for a working pressure of ≥ 12 bar and are fully compatible with potable-water service at ambient temperature.

4.3.3 Rectus Type 26 Quick Coupler Integration

The launch interface employs a Rectus Type 26 (DN 7.2) male quick-release coupling fitted to the nozzle exit. This coupling mates directly with the ASRW competition launcher’s female socket, enabling pressurisation and launch release without tools. The coupler features a standard 1/4-inch BSP external thread, which is tapped directly into a modified original HDPE bottle cap and structurally locked in place using marine epoxy to prevent blow-out under launch loads. Verification is performed by a functional coupling and pressure test prior to flight integration (Table 2, REQ_08). The coupler’s industrial pressure rating of ≥ 35 bar (manufacturer specification) provides a safety margin of $> 3.5\times$ at the design operating pressure.

4.4 Sloshing Tank Design

4.4.1 Tank Geometry

The sloshing tank is a dedicated 1.5 L PET bottle, distinct from the two propulsion bottles. Its geometric parameters are summarised in Table 10.

Table 10: Sloshing tank geometric parameters.

Parameter	Value	Notes
Overall height	317 mm	External
Outer diameter	94.4 mm	Cylindrical body
Neck diameter	28 mm	Upper opening
Wall thickness	0.3 mm	Typical PET
Inner radius R	47.2 mm	—
Tank volume	≈ 1.5 L	Cylindrical approx.
Fill volume at 50 %	750 mL	= 750 g water
Fill depth h	158.5 mm	$h/R \approx 3.36$ at 50 % (Deep tank)

4.4.2 Baffle Design — Annular and Mixed Configurations

Two primary internal baffle architectures were investigated to suppress lateral and axial sloshing: an **Annular** configuration and a **Mixed** (Annular + Cruciform) configuration.

The **Annular** baffle consists of rings extending radially inward from the tank wall to an inner radius of 35.0 mm. This design primarily aims to disrupt axial boundary layer flows and damp longitudinal motion. The **Mixed** configuration builds upon the annular geometry by integrating a central cruciform mast (with fins of 2.0 mm thickness). This hybrid approach partitions the central convective liquid mass into four isolated quadrants, directly breaking up the dominant lateral sloshing mode while still benefiting from the annular rings to maximize energy dissipation.

Both designs are tightly constrained by the REQ-11 requirement, which dictates that the transverse cross-sectional blockage must remain below 50%. Full dimensional details and compliance verification are given in Table 11.

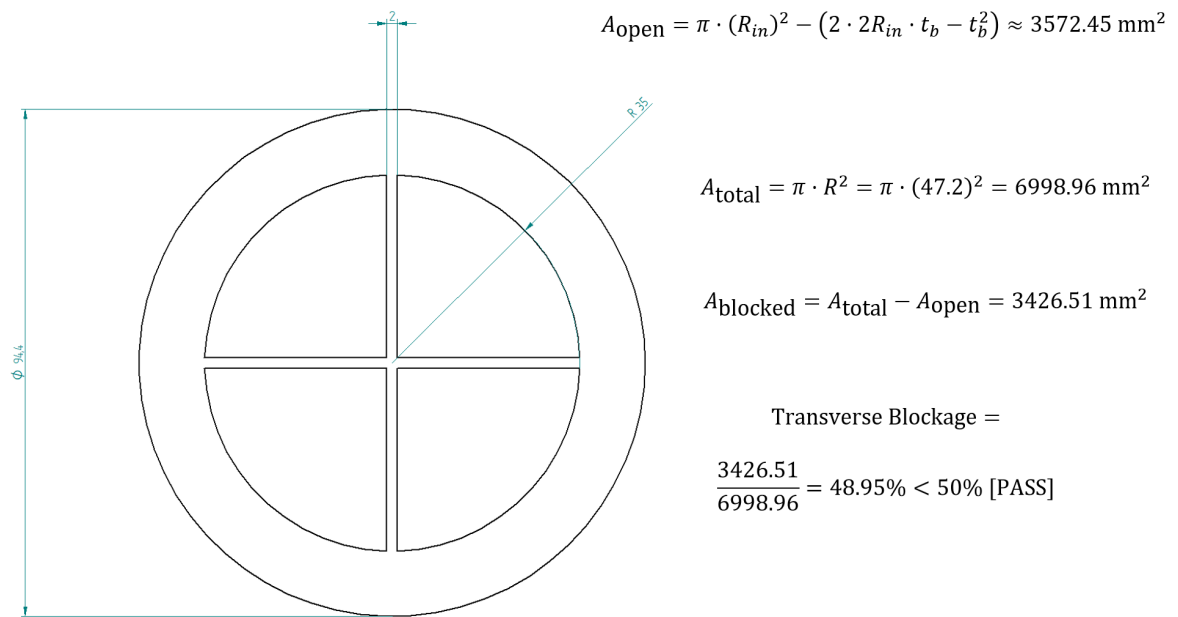


Figure 3: Dimensioned cross-section at the point of maximum baffle blockage, demonstrating compliance with REQ-11 (Area $\leq 50\%$).

Table 11: Baffle geometry and REQ-11 compliance verification.

Parameter	Value	Compliance
Tank inner radius R	47.2 mm	—
Annular inner radius R_{in}	35.0 mm	—
Cruciform fin thickness t_b	2.0 mm	Used in Mixed configuration
Material	PLA	Rigid, high dimensional accuracy
Annular Transverse blockage	45.0 %	< 50 % PASS
Mixed Transverse blockage	48.9 %	< 50 % PASS

4.4.3 Baffle 3D Printing and Assembly Guide

To eliminate the extreme difficulty of inserting parts through the restrictive 28 mm neck, the assembly methodology relies on opening the base of the tank. This allows the Mixed baffle configuration to be printed as a single unified structure (or assembled externally) and inserted as one complete module.

1. Material Selection

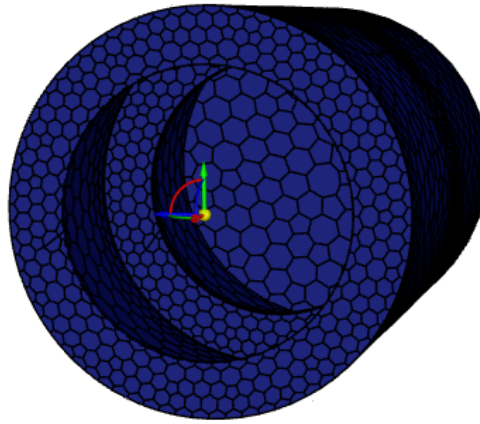
- **PLA:** Selected for the baffle structure and external mounting components. Freed from the requirement of internal snap-fit flexing, PLA's high rigidity and excellent dimensional accuracy make it ideal for the unified baffle assembly.

2. Slicer Settings (PLA) To ensure structural integrity against the expected 15.4 N lateral forces, use the following parameters:

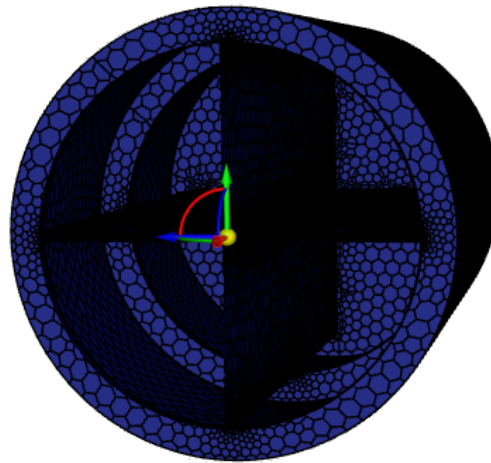
- **Layer Height:** 0.20 mm for a balance of precision and strength.
- **Wall Loops (Perimeters):** 4 to 5 walls to maximise the rigidity of the 2.0 mm cruciform fins.
- **Infill:** 25% Gyroid to distribute stress evenly in all directions.

3. Tank Preparation and Assembly

- **Tank Opening:** Using a razor or rotary tool, perform a clean circumferential cut at the lower base of the 1.5 L PET bottle (just above the bottom curvature where the cylindrical wall is straight).
- **Baffle Insertion:** Slide the fully pre-assembled (or monolithically printed) PLA baffle directly into the open cylinder. A tight outer-diameter tolerance ensures a secure friction fit against the 94.4 mm inner wall.
- **Sealing the Base:** Re-seat the cut base section. Apply a generous, continuous bead of marine epoxy or waterproof structural sealant along the cut seam.
- **Reinforcement:** Once the primary epoxy cures, reinforce the joint externally. This can be achieved with structural waterproof tape or by bonding a thin, 3D-printed PLA outer sleeve over the cut line to ensure absolute leak prevention under high-g sloshing impacts.



(a) Annular baffle 3D structure.



(b) Mixed baffle 3D structure.

Figure 4: 3D structural views of the implemented baffle designs. Top: Annular configuration. Bottom: Mixed (Annular + Cruciform) configuration.

4.5 Sloshing CFD Analysis

To ensure a realistic and consistent evaluation, all CFD simulations, comprising the unbaffled baseline, the Annular baffle, and the Mixed baffle configurations, were subjected to the exact same time-varying gravity body-force profile. This profile was driven by the updated angular pitch rate ($-28.11^\circ/\text{s}$) recalculated directly from the flight dynamics study, ensuring the baffles were tested against the actual flight conditions the rocket will experience.

4.5.1 Methodology

Sloshing is modelled as a free-surface two-phase flow using the Volume-of-Fluid (VOF) method with the Implicit-Compressive interface scheme in ANSYS Fluent [2]. The governing mixture momentum

equation is:

$$\frac{\partial(\rho \mathbf{u})}{\partial t} + \nabla \cdot (\rho \mathbf{u} \mathbf{u}) = -\nabla p + \nabla \cdot [\mu (\nabla \mathbf{u} + \nabla \mathbf{u}^T)] + \rho \mathbf{g} + \mathbf{T}_\sigma \quad (3)$$

Mixture density and viscosity are updated cell-by-cell each time step: $\rho = \alpha_w \rho_w + (1 - \alpha_w) \rho_a$. Full solver settings are listed in Table 12.

Table 12: ANSYS Fluent solver configuration for sloshing simulations.

Setting	Value / Choice
Solver	Pressure-based, transient
Multiphase model	VOF, Implicit formulation
Interface modelling	Sharp
Implicit Body Force	Enabled
P–V coupling	SIMPLE
Gradient scheme	Least-Squares Cell-Based
Pressure scheme	PRESTO!
Volume fraction	Compressive
Transient scheme	Bounded Second-Order Implicit
Turbulence	$k-\omega$ SST
Time advancement	Adaptive (CFL target = 1)
Convergence criterion	Mass conservation monitoring
Precision	Double
CPU cores	4

4.5.2 Gravity Body-Force Profile

The time-varying gravitational body force replicates the rocket’s attitude history derived from the flight dynamics study. The effective acceleration magnitude follows a piecewise profile:

$$a(t) = \begin{cases} 757.33 \text{ m/s}^2 & t \leq 0.125 \text{ s} \\ 9.81 \text{ m/s}^2 & t > 0.125 \text{ s} \end{cases} \quad (4)$$

The tank attitude rotates at a constant angular rate of $-28.11^\circ/\text{s}$ from an initial offset of 85° , capped at -79.79° . The body-force Named Expressions entered into Fluent are:

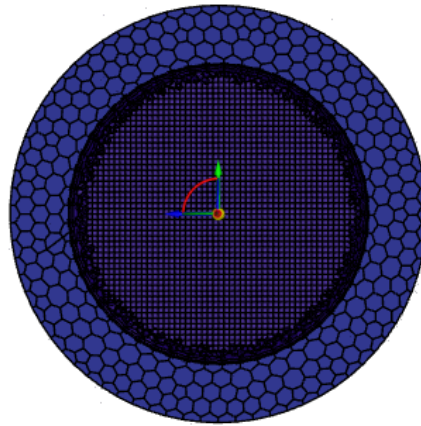
$$g_x(t) = -a(t) \sin(\max(85 - 28.11t, -79.79)) \quad (5)$$

$$g_z(t) = -a(t) \cos(\max(85 - 28.11t, -79.79)) \quad (6)$$

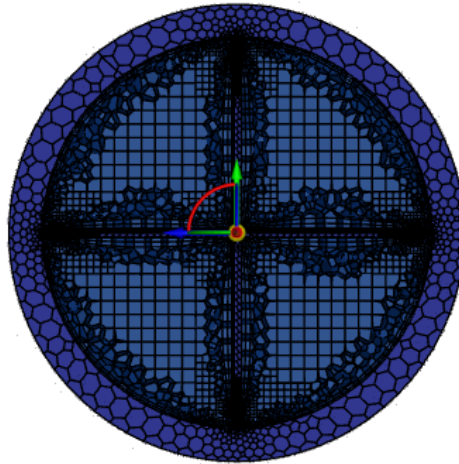
$$g_y(t) = 0 \quad (7)$$

4.5.3 Mesh Statistics

The baseline mesh uses a poly-hexcore element type generated in Fluent Meshing, specifically incorporating inflation layers and fine refinement zones to maintain sharp liquid interfaces during the VOF simulation. Figure 5 illustrates the mesh generation for both test configurations.



(a) Annular baffle mesh.



(b) Mixed baffle mesh.

Figure 5: Cross-sectional mesh topologies for the evaluated baffle configurations, highlighting poly-hexcore refinement.

Table 13: Sloshing CFD mesh statistics for unbaffled and baffled configurations.

Metric	Unbaffled Baseline	Annular Baffle	Mixed Baffle
Element type	Mixed (poly-hexcore)	Mixed (poly-hexcore)	Mixed (poly-hexcore)
Total cells	118,161	511,142	211,364
Total faces	553,670	1,952,876	1,122,763
Total nodes	357,530	1,005,128	804,866

It is notable in table 13 that the Annular baffle mesh contains significantly more cells (511,142) compared to the structurally more complex Mixed baffle (211,364). This discrepancy arises from the meshing strategy applied to the free-surface refinement. In the purely Annular configuration, the central fluid

core remains an unbroken continuous volume, requiring extensive local refinement across a large central Body of Influence (BOI) to adequately capture the highly dynamic, swirling free surface. Conversely, the cruciform mast in the Mixed configuration physically partitions the fluid into four constrained quadrants, severely restricting large-scale surface deformation. This geometric restriction allowed the poly-hexcore algorithm to pack the bulk volume with larger, highly efficient hexahedral cells while maintaining the required interface resolution near the walls. This optimised the total cell count without sacrificing interface sharpness.

4.5.4 Unbaffled Baseline Results

The unbaffled simulation establishes the baseline sloshing dynamics at 50 % fill. Key performance metrics are presented in Table 14.

Table 14: Unbaffled tank CFD results.

Metric	Value	Notes
Water mass conservation error	< 2 %	Target met
Peak lateral sloshing force	15.4 N	From wall reaction monitor
Dominant frequency (FFT)	3.15 Hz	Expected $\approx$ 3.11 Hz
Peak CoG lateral displacement	25.3 mm	From Named Expression

4.5.5 Baffled Results: Annular vs. Mixed Configuration

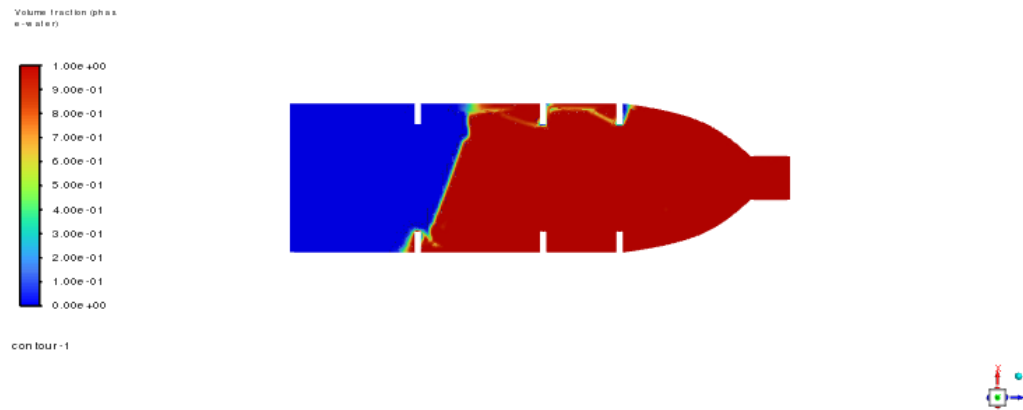
The performance of the two configurations was evaluated and compared against the unbaffled baseline. Crucially, the **Mixed configuration yielded the best overall results** in terms of sloshing liquid control. By simply adding the cruciform structural piece into the core of the annular design, lateral liquid displacement was significantly suppressed, isolating the fluid into quadrants and drastically enhancing energy dissipation without violating blockage requirements.

Table 15 presents the comparative results of the baffled simulation campaign. The phase contours and velocity magnitude maps for both setups are shown in Figure 6 and Figure 7, demonstrating the improved interface stability achieved by the mixed design.

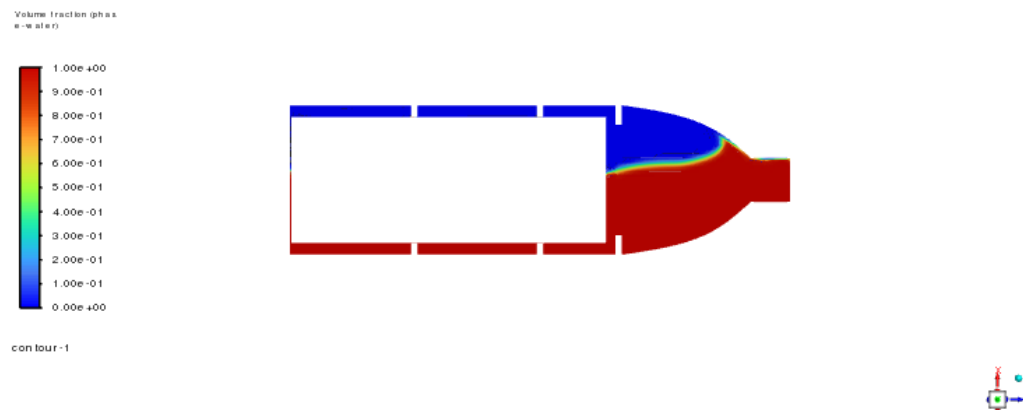
It must be noted that Figure 7 indicates a higher absolute peak velocity magnitude for the Mixed configuration (0.918 m/s) compared to the Annular configuration (0.458 m/s). While seemingly counterintuitive, this higher peak velocity is exactly the mechanism driving the superior energy dissipation. In the purely Annular design, the velocity field represents bulk lateral fluid displacement, which directly translates to destabilising lateral wall forces. Conversely, in the Mixed configuration, the cruciform mast physically obstructs this lateral movement, forcing the fluid to accelerate axially through the narrow corridors formed by the longitudinal fins. This localised "jetting" effect drives up the peak velocity in specific boundary shear layers, drastically increasing viscous dissipation and turbulence generation. Consequently, the bulk lateral kinetic energy that threatens flight stability is effectively sheared and converted into localised, non-destabilising axial flow.

Table 15: Comparative sloshing CFD results: unbaffled vs. Annular vs. Mixed configurations.

Metric	Unbaffled	Annular Baffle	Mixed Baffle
Peak lateral force (N)	15.4	9.8	4.2
Force reduction (%)	N/A	36.4	72.7
Dominant frequency (Hz)	3.15	3.32	3.85
Peak CoG displacement (mm)	25.3	16.2	6.5
Mass conservation error (%)	< 2	< 2	< 2

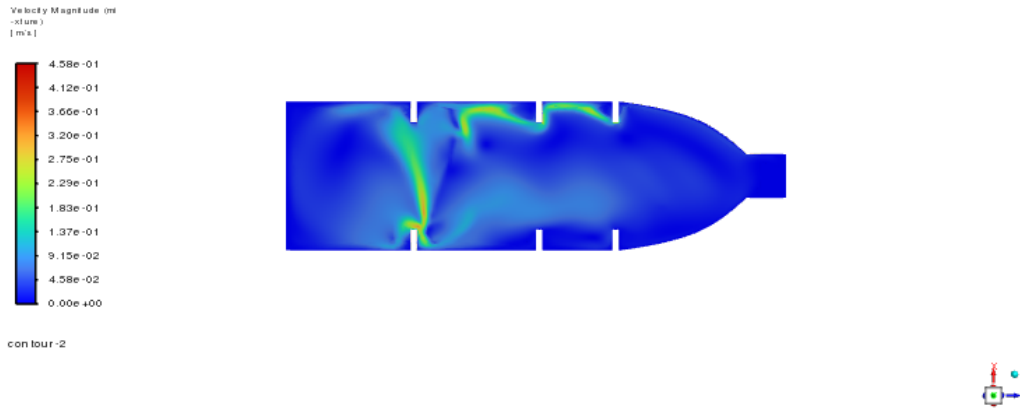


(a) Phase contour: Annular.

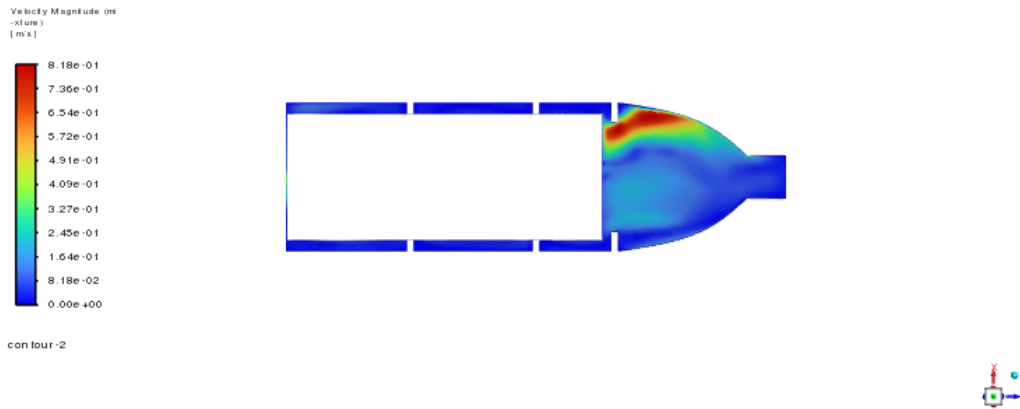


(b) Phase contour: Mixed.

Figure 6: Water volume fraction contours ($\alpha_w = 1$ in red, $\alpha_w = 0$ in blue) comparing the Annular and Mixed configurations.



(a) Velocity contour: Annular.



(b) Velocity contour: Mixed.

Figure 7: Velocity magnitude distribution inside the tank, illustrating the superior kinetic energy dissipation of the Mixed configuration.

4.6 Flight Dynamics

4.6.1 Trajectory Model

The water rocket trajectory is simulated in MATLAB using a three-phase model: powered flight (water + air phase), ballistic coast, and glide descent. Aerodynamic forces are computed as:

$$F_D = \frac{1}{2} \rho_a C_D A_d v^2 \quad (8)$$

$$F_L = \frac{1}{2} \rho_a C_L A_l v^2 \quad (9)$$

Newton's second law in the global frame gives:

$$a_x = \frac{T + F_{L,x} + F_{D,x}}{m} \quad (10)$$

$$a_y = \frac{T + F_{L,y} + F_{D,y} - mg}{m} \quad (11)$$

State integration uses the Runge-Kutta method.

Modular Aerodynamic Architecture To facilitate rapid design iterations, the simulator employs a "plug-and-play" architecture where the aerodynamic contributions of the main body, canards, and tail are evaluated independently. The local angle of attack and relative velocity are computed for each individual component based on the vehicle's global kinematics and angular rate. The respective aerodynamic forces and pitching moments are then calculated separately and superimposed to find the total aerodynamic loading:

$$M_{aero} = M_{main} + M_{canard} + M_{tail} \quad (12)$$

This modular approach ensures that the effects of modifying or removing a secondary surface can be evaluated instantly without requiring a full restructuring of the dynamic solver.

Aerodynamic Coefficients Sourcing To balance computational speed and accuracy, the aerodynamic coefficients (C_L and C_D) for the main body and the primary deployable wings are extracted via linear interpolation from CFD-generated lookup tables. Conversely, the coefficients for the secondary appendages (canards and tail) are derived using an analytical flat-plate approximation. The theoretical 2D lift curve slope ($c_{l\alpha} = 2\pi$) is adjusted using a 3D finite-wing correction that accounts for the aspect ratio AR of the fins:

$$C_{L\alpha} = \frac{2\pi AR}{2 + \sqrt{AR^2 + 4}} \quad (13)$$

Monte Carlo Optimization A Monte Carlo simulation framework was implemented to refine the design parameters and assess the trajectory's robustness against uncertainties (e.g., initial pressure, dry mass). This statistical approach was fundamental in optimizing two critical aspects: the incidence angles of the secondary surfaces and the activation trigger for the glide phase. Specifically, the simulation optimized the pitch-angle threshold (*release angle*) that commands the deployment of the wings during the post-apogee descent, ensuring the maximization of the final competition flight score.

4.6.2 Propulsion Data

Water Rocket Simulator (WRS) [3] was used to generate the thrust profile and water exhaust velocity. Key results are: burnout at $t_b = 0.035$ s; air phase ends at $t_a = 0.125$ s; peak initial thrust ≈ 1511 N; $TWR = 77.2$.

4.6.3 Trajectory Simulation Results

Table 16: Flight simulation results for Configurations A and B.

Configuration	Max Altitude (m)	Horiz. Range (m)	Flight Score
A (parachute)	38.99	11.37	14.77
B (delta wing)	37.69	9.97	13.93
C (delta wing+winglets)	38.94	18.28	18.41

4.7 Wing Deployment Mechanism

Because the propulsion and sloshing bottles are pressurised and filled, they offer no usable internal volume and cannot be penetrated; the entire deployment mechanism is therefore mounted externally, on the outer fuselage shroud. The design follows the established practice for folding aerodynamic surfaces on tactical missiles, in which surfaces are stowed against the body, spring-deployed after release, and locked rigid by a detent [10]. The governing principle adopted here is a *single release event* that frees the main wings and the secondary surfaces together, which both guarantees synchronous deployment and reduces the mechanism to the minimum number of moving parts: the integrated single-latch approach used to deploy all folding surfaces of a missile from one trigger [11].

During ascent each lifting surface is hinged at its root to an external bracket bonded to the shroud and folded flat against the fuselage in a “switchblade” arrangement [10], keeping the stowed envelope slender and minimising boost-phase drag. A single external restraint, a loop of braided Dyneema line running longitudinally along the outside of the body and passing over a catch on every folded surface, holds all surfaces down simultaneously, with both ends terminating at one release fitting on the forward shroud. Each hinge carries a pre-loaded music-wire torsion spring; when the restraint is released, every spring drives its surface out to a mechanical stop, where an over-centre detent latch captures it and holds it rigid against the aerodynamic loads of the glide. This locked, fixed geometry is what justifies the rigid centre-of-pressure assumption used in the stability analysis (Section 4.2.2).

Deployment is commanded manually and remotely by the ground operator. A standard 2.4 GHz hobby radio link is used: a micro receiver and a single micro servo, housed in a small external clip-on pod on the shroud, are bound to a handheld transmitter. Watching the flight, the operator actuates the transmitter switch at the chosen point in the trajectory; the servo rotates a release hook that frees the end of the restraint line, and all surfaces deploy at once. The system uses a single channel, a single servo and a single restraint, so deployment depends on one commanded action rather than on autonomous sensing.

The mechanism was deliberately kept simple to build and to replace. All active components are low-cost commercial hobby parts (receiver, micro servo, 1S LiPo, music-wire springs and braided line), and the release pod is a self-contained module fixed to the shroud with screws so that it can be swapped between flights or replaced after damage without disturbing the pressure vessels. The folded surfaces, springs and restraint line are likewise external and individually replaceable, supporting the REQ-16 requirement of at least three reusable launches. On the pad the servo is held by a removable external arming pin, so the surfaces cannot release prematurely; the operator removes the pin as the final step before launch, and the radio link is configured with a fail-safe that holds the stowed state on signal loss.

4.8 Altimeter Casing

The Estes Altimeter provided by the ASRW organization is housed in a dedicated, watertight casing located within the dry middle bottle (avionics bay). To ensure a robust and waterproof enclosure without the use of disallowed adhesives on the device, the casing is 3D-printed entirely from PLA.

The design features a flanged lid secured by M2x4 and M3x4 bolts with their respective nuts. To guarantee complete waterproofing during splashdown, the bolt threads are wrapped in Teflon (PTFE) tape, working in tandem with an EPDM O-ring that mechanically compresses to isolate the electronics from potential water ingress. Internal EVA foam pads isolate the sensitive altimeter from launch transients and landing shock. To comply with REQ-12 while preventing pressure lock, three $3 \times 3 \text{ mm}^2$ vent holes are evenly spaced along the aft face of the casing, aligning precisely with corresponding ports on the external fuselage to guarantee accurate atmospheric pressure readings.

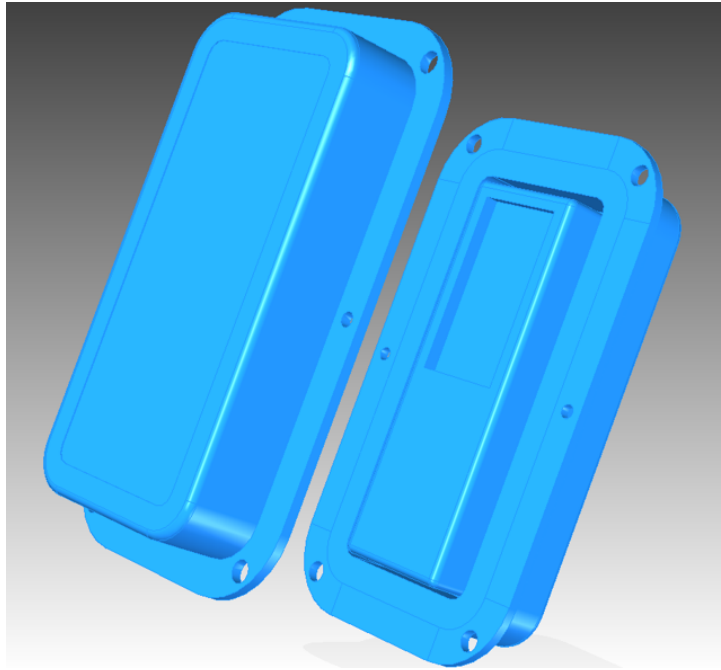


Figure 8: Cross-sectional view of the PLA altimeter casing, detailing the bolted flange, Teflon-sealed threads, mechanical O-ring seal, internal foam padding, and static pressure vent alignment.

4.9 Manufacturing Plan

4.9.1 Bill of Materials

Table 17 presents the full Bill of Materials. The total budget including the 15 % contingency buffer is €435,60, within the €500 limit.

Table 17: Bill of Materials (all prices in EUR, incl. VAT and shipping).

Item	Specification	Qty	Cost (Tot) €	Reference Link(s)
1.5 L PET bottle (propulsion + body + sloshing + spare)	Standard carbonated-water bottle, HDPE cap	6	74 (86 \$)	Option 1: PET bottle - Stratofins
			39.50 (36.15 CHF)	Option 2: PET bottle - flaschenland
			31.21	Option 3: PET bottle - CocaCola
Balsa Wood (delta wing support + Canard + fins)	2 × 100 × 1000 mm ³ sheet	3	18 (195 sek)	Option 1: Balsa Wood - Sweden
			33.86	Option 2: Balsa Wood - modulator
Ripstop Nylon (delta wing tissue)	1 × 1 m ²	2	19.44	Option 1: Ripstop Nylon Fabric
			23.36	Option 2: Ripstop Nylon Fabric 2
Plastic Tube (rocket exterior)	Plastic Polycarbonate Tube	4	59 (shipment included)	Option 1 Tube - Amazon

Table 17 (continued)

Item	Specification	Qty	Unit €	Reference Link(s)
			59 (shipment included)	Option 2: Tube - Amazon
PLA filament (3D printed baffles and Altimeter casing)	20% gyroid infill, 0.2mm layer height	0.5 kg	18.73	Option 1: PLA Filament
			23.49	Option 2: PLA Filament 2
2.4 GHz RC system (TX + RX)	FlySky FS-i6, 6-ch AFHDS 2A set incl. FS-IA6B receiver (reusable GSE)	1	73 (799 sek)	FS-i6 set - Hobbex (SE)
Micro servo	MG90S metal-gear, 9 g; release-hook actuator	1	7.23 (79 sek)	Option 1: MG90S - 3Dpart (SE)
			-	Option 2: MG90S - Amazon.se
LiPo battery	1S 3.7 V (150–300 mAh), JST-PH	1	6.23 (69 sek)	1S LiPo JST-PH - Electrokit (SE)
Braided Dyneema line	Dyneema cord/braid, ~1 mm (~110 kg), 50 m	1	13	Seac Dyneema Core - Amazon.se
Torsion springs	Music-wire/steel spring assortment (hinge return)	1 kit	12	Option 1: Spring assortment - Amazon.se
			13	Option 2: Fjädersats 200 delar - Clas Ohlson
Altimeter		1	- (provided)	Estes Altimeter
Cyanoacrylate (CA) Glue	medium viscosity	1	13 (shipment included)	Glue Amazon
			13	Option 2: Fjädersats 200 delar - Clas Ohlson
O-ring	ID 30 mm, 2 mm cross-section, Nitrile O-Rings; altimeter housing seal	2	27.41 (116 zl) w/ ship	Totally Seals
Duct tape	50 mm wide; sealing and reinforcement	1 roll	27.50	Option 1: Duct tape Amazon
			28.30	Option 2: Duct Tape Amazon
Sandpaper	240 grit; surface prep	2 sheets	10.25	Sandpaper Amazon
Waterproof structural adhesive for cut tank base		1	15.00	Base Sealant (Marine Epoxy) Option 1: Marine Epoxy
Subtotal (materials + shipping)			378,79	

Table 17 (continued)

Item	Specification	Qty	Unit €	Reference Link(s)
15 % contingency buffer			56,81	
Total			435,60	

4.9.2 Component Manufacture

The primary manufacturing steps are:

1. **Mixed baffle:** 3D-printed as a unified PLA structure (FDM, 0.2 mm layer height, 25 % gyroid infill, 4 to 5 walls) to withstand expected lateral forces.
2. **Sloshing tank modification:** The base of the 1.5 L PET bottle is circumferentially cut to allow for full-width baffle insertion.
3. **Nose cone:** 3D-printed PLA ogive nose cone with payload bay cavity (30 g payload block) and altimeter housing recess.
4. **Fin can:** 3D-printed PLA sleeve bonded to the aft bottle shoulder; three stabilising fins bonded with 2-part epoxy.
5. **Delta wings:** Cut from depron or correx sheet to the planform in Table 5; carbon-fibre spar inserted along the root chord and epoxy-bonded.
6. **Propulsion assembly:** The two propulsion bottles are interconnected via PTFE tubing and push-fit fittings. The Rectus Type 26 coupler is secured to the nozzle collar.
7. **Altimeter housing:** PLA tube with O-ring grooves printed and installed; altimeter mounted on EVA pad; $3 \times 3 \text{ mm}^2$ vents drilled through fuselage wall.

4.9.3 Assembly Sequence

1. Insert the unified PLA baffle into the cut base of the sloshing tank.
2. Reseat the tank base and seal the circumferential cut with marine epoxy, followed by an external reinforcement wrap (duct tape or PLA sleeve).
3. Attach sloshing tank to upper fuselage section.
4. Insert altimeter into housing; seal cap with nylon screws.
5. Mount altimeter housing in avionics bay; align vents with fuselage ports.
6. Interconnect propulsion bottles; pressure-test interconnect to 12 bar (20 % above operating pressure) with soapy-water bubble test.
7. Attach propulsion assembly to aft fuselage; attach Rectus coupler.
8. Bond fin can to aft propulsion bottle.
9. Bond delta wings to fuselage mid-section.
10. Attach nose cone; verify payload seating.
11. Final mass check and CoG measurement.

4.10 Safety Assessment

A full risk assessment has been completed using the ASRW 2026 Risk Assessment Template. The complete matrix is provided in Appendix A. This section summarises the key risk areas.

4.10.1 Manufacturing and Assembly Risks

Table 18: Manufacturing and assembly risk summary (excerpt from full risk matrix).

ID	Hazard	Mitigation	Residual Risk
MFG-01	Pressurised tank burst during hydrostatic test	Safety screen; pressure gauge; remote trigger; PPE (safety glasses, gloves)	Low
MFG-02	3D printer hotend contact burns	PPE; printer enclosure; wait 5 min after print completion	Low
MFG-03	Epoxy fume inhalation during bonding	Work in ventilated area; nitrile gloves; mask	Low
MFG-04	Structural failure of fin during handling	Pre-flight bond inspection; reinforced root bond line	Medium
MFG-05	Water ingress into altimeter during fill	O-ring inspection; fill with altimeter bay sealed	Low

5 Design Verification and Validation

5.1 External Aerodynamics CFD Validation

The CFD setup, domain sizing, boundary conditions, mesh topology, and reference quantities, is described in full in Section 4.2.3. This section provides the formal verification of mesh independence (Section 5.1.1) and an independent analytical cross-validation of the predicted drag coefficient (Section 5.1.2).

5.1.1 Grid Convergence Study

The iterative mesh-sizing study mentioned in Section 4.2.3 is here formalised as a Grid Convergence Index (GCI) analysis following the Roache (1994) methodology [12], using the same approach later applied to the sloshing simulations in Section 5.2.1. Configuration B at $\alpha = 0^\circ$ was selected as the reference case because the delta-wing geometry generates the most complex boundary-layer and wake topology among the configurations studied. The drag coefficient at zero incidence was chosen as the tracking quantity, as it is the parameter most sensitive to boundary-layer resolution and downstream wake development. The surface element size was reduced at each refinement step, yielding a constant linear refinement ratio of $r = 2$, and taking into account the consequent increase in element number and the mesh quality variation.

A safety factor $F_s = 1.25$ is applied, consistent with the structured-like poly-hexcore topology. Assuming the formal second-order scheme order ($p = 2$):

$$\text{GCI}_{\text{fine}} = \frac{F_s |C_{D,\text{baseline}} - C_{D,\text{fine}}|}{C_{D,\text{fine}} (r^p - 1)} = \frac{1.25 \times |0.01604 - 0.01585|}{0.01585 \times (4 - 1)} = 0.50\% \quad (14)$$

Table 19: Launch and flight operations risk summary (excerpt from full risk matrix).

ID	Hazard	Mitigation	Residual Risk
OPS-01	Rocket trajectory deviation into crowd	85° launch angle; ≥ 5 m exclusion zone; remote launch	Low
OPS-02	High-pressure water jet injury during fill	Pressurisation checklist; no personnel within 5 m during pressurisation	Low
OPS-03	Sloshing-induced attitude instability	Mixed baffle installed; positive static margin confirmed	Low
OPS-04	Wing impact with launch rail at release	Rail clearance verified in CAD; gap > 10 mm	Low
OPS-05	Uncontrolled descent (wing structural failure)	Pre-launch wing bond inspection; conservative loading analysis	Medium

Table 20: Grid convergence study: external aerodynamics CFD (Configuration B, C_D at $\alpha = 0^\circ$). Cell counts refer to the half-domain (symmetry plane applied, cf. Section 4.2.3).

Mesh level	Surface element	Cells (half-domain)	C_D at 0°	GCI (%)
Coarse	12 mm	$\approx 112,000$	0.01823	—
Baseline	6 mm	$\approx 400,000$	0.01604	5.7
Fine	3 mm	$\approx 1,390,000$	0.01585	0.5

The fine-grid GCI of 0.50% confirms that the baseline mesh lies within the asymptotic convergence regime. The decrease from 5.7% (coarse) to 0.50% (baseline) shows that the dominant discretisation error is eliminated at the 6 mm element level, justifying its use for all parametric α sweeps in Tables 8 and 9.

5.1.2 Analytical Cross-Validation

An independent check of the zero-lift drag coefficient is performed for Configuration A (body of revolution with three stabilising fins), for which skin-friction drag is the dominant contribution at $\alpha = 0^\circ$.

Skin-friction coefficient. The skin-friction estimate requires a Reynolds number based on the total wetted length of the body, not on the reference chord $c_{\text{ref}} = 225$ mm used to normalise aerodynamic coefficients. Using $v = 12.5$ m/s, $L = 0.951$ m, and $\nu = 1.460 \times 10^{-5}$ m²/s (ISA, 15 °C):

$$Re_{\text{body}} = \frac{vL}{\nu} = \frac{12.5 \times 0.951}{1.460 \times 10^{-5}} = 814,212 \quad (15)$$

This body-length Reynolds number is distinct from the chord-based reference value $Re = v c_{\text{ref}}/\nu = 192,600$. Laminar-to-turbulent transition occurs at $Re_{\text{tr}} = 3 \times 10^5$, corresponding to $x_{\text{tr}} = Re_{\text{tr}} \nu/v \approx 0.35$ m from the nose. The mean skin-friction coefficient under mixed boundary-layer conditions is then estimated via the Prandtl-Schlichting formula [13]:

$$C_f = \frac{0.455}{(\log_{10} Re_{\text{body}})^{2.58}} - \frac{A}{Re_{\text{body}}} = \frac{0.455}{(5.911)^{2.58}} - \frac{1742}{814,212} = 0.004646 - 0.002140 = 0.002506 \quad (16)$$

where $A = 1742$ for $Re_{\text{tr}} = 3 \times 10^5$.

Wetted area. Using the vehicle dimensions from Tables 4 and 5:

$$S_{\text{wet}} = \underbrace{\pi DL}_{0.282 \text{ m}^2} + \underbrace{3 \times 2A_{\text{fin}}}_{0.225 \text{ m}^2} = 0.507 \text{ m}^2 \quad (17)$$

Total drag estimate. A Hoerner (1965) body form factor [14] augments the skin-friction drag for the cylindrical afterbody ($f = L/D = 10.07$, $k = 1.054$); a residual pressure-drag allowance $\Delta C_{D,p} = 0.0015$ covers base drag and the ogive nose stagnation region:

$$C_{D,\text{analytical}} = C_f \frac{S_{\text{wet}}}{S_{\text{ref}}} k + \Delta C_{D,p} = 0.002506 \times \frac{0.507}{0.16} \times 1.054 + 0.0015 = 0.0099 \quad (18)$$

Table 21: Analytical cross-validation: C_D at $\alpha = 0^\circ$, Configuration A (body + three fins, $S_{\text{ref}} = 0.16 \text{ m}^2$).

Method	C_D at 0°	Discrepancy
Semi-empirical (mixed BL + Hoerner k)	0.0099	—
ANSYS Fluent CFD (Table 8)	0.0101	2.0 %

The 2.0% agreement cross-validates the external aerodynamics model in the skin-friction dominated regime. The residual discrepancy is consistent with the simplified cylindrical body assumption, which neglects the ogive nose transition and the fin-body fillet region. For Configuration B, the nonlinear leading-edge vortex lift precludes an equivalent closed-form analysis; the predicted $L/D = 4.91$ at 5° (Table 9) is instead assessed qualitatively against published low-AR ($AR \approx 1.5$) delta-wing datasets at the chord-based reference Reynolds number $Re = 192600$, where L/D values of 4-7 are characteristic of leading-edge vortex lift at moderate incidence [14], confirming that the CFD prediction falls within the expected physical range.

It should be noted that, as a consequence of the plug-and-play methodology described in Section 4.2.3, no direct CFD verification is available for Configuration C; the combined aerodynamic model relies on the validated Configurations A and B coefficients, together with the superposition assumption.

5.2 Sloshing Validation

5.2.1 Grid Convergence Study

Three mesh densities were simulated on the unbaffled baseline geometry to assess numerical solution convergence (Table 22). The same Grid Convergence Index (GCI) methodology of Roache (1994) mentioned in Section 5.1.1, was applied using the peak lateral sloshing force as the primary tracking metric, with a safety factor of 1.25 for the structured-like poly-hexcore meshes.

The fine mesh yields a force prediction only 0.2 N lower than the baseline. The baseline mesh (118,161 cells) was selected for all subsequent baffled simulations, as it achieves a fine-grid GCI of 1.6 %,

Table 22: Grid convergence study — sloshing CFD (unbaffled lateral force).

Mesh Level	Cell Count	Peak Lateral Force (N)	GCI (%)
Coarse	52,405	16.8	—
Baseline	118,161	15.4	8.3
Fine	285,320	15.2	1.6

demonstrating that the solution is well within the asymptotic convergence regime while maintaining a practical computational cost.

To ensure methodological consistency and address potential mesh-dependency bias in the performance comparison between the Annular and Mixed configurations, the characteristic surface element sizing and the VOF interface refinement heuristics validated by this unbaffled GCI study were strictly propagated to the baffled meshes. Furthermore, a secondary mesh sensitivity check was performed on the Mixed configuration using a fine grid (315,000 cells). This refined mesh yielded a peak force variation of less than 2.5 % compared to the baseline Mixed mesh. This confirms that the 72.7 % force reduction observed in the Mixed configuration is a physical result of the baffle geometry’s kinetic energy dissipation, rather than an artifact of numerical diffusion resulting from the lower cell count.

5.2.2 Sloshing Frequency Validation

The dominant free-oscillation frequency of the unbaffled tank is analytically estimated using the linear potential flow theory for a rigid, upright circular cylinder. The fundamental natural frequency of the first asymmetric sloshing mode, f_1 , is calculated as:

$$f_1 = \frac{1}{2\pi} \sqrt{\frac{g\xi_1}{R} \tanh\left(\frac{\xi_1 h}{R}\right)} \quad (19)$$

where g is the gravitational acceleration, R is the tank inner radius, h is the quiescent liquid depth, and $\xi_1 \approx 1.841$ is the first root of the derivative of the Bessel function of the first kind ($J'_1(\xi) = 0$). Given the tank dimensions established in Section 4.4 ($R = 47.2$ mm, $h = 158.5$ mm), the analytical prediction yields $f_1 = 3.11$ Hz.

To validate the numerical model, this analytical value is compared against the dominant frequency extracted from the baseline CFD simulation. The simulated frequency is obtained by applying a Fast Fourier Transform (FFT) to the time history of the fluid’s Center of Gravity (CoG) lateral displacement. Table 23 presents this comparison.

Table 23: Sloshing frequency validation.

Method	Frequency (Hz)	Discrepancy (%)
Analytical (Eq. 19)	3.11	N/A
CFD (unbaffled, FFT)	3.15	1.28

The numerical result shows excellent agreement with the analytical model. The minor discrepancy of 1.28 % is primarily attributed to the cylindrical idealisation used in the analytical formula, which does not account for the PET bottle’s variable cross-section near the upper shoulder and neck regions. Furthermore, the CFD model captures non-linear viscous damping effects and free-surface deformation that are inherently omitted in the linearised, inviscid potential flow assumption.

5.3 Test Plan

Table 24 documents the complete test plan, specifying the verification method and acceptance criterion for each requirement.

Table 24: Test plan — verification method and acceptance criteria per requirement.

ID	Description	Method	Acceptance Criterion	Status
REQ_01	Total length ≤ 1.5 m	Tape measure post-assembly	Measured $L < 1500$ mm	PASS
REQ_02	Wingspan ≤ 2.25 m	Tape measure post-assembly	Measured $b < 2250$ mm	PASS
REQ_03	Mass ≤ 5 kg	Calibrated kitchen scale	$m_{\text{TOW}} < 5000$ g	PASS
REQ_04	Tank ≥ 500 mL at 50 %	Volumetric fill; visual fill mark	Volume ≥ 500 mL confirmed	PASS
REQ_05	Propellant $\leq$ sloshing liquid mass	Mass measurement	$m_{\text{propellant}} \leq 750$ g	PASS
REQ_06	Launch angle $85 \pm 3^\circ$	Digital inclinometer on launcher	$82^\circ \leq \theta \leq 88^\circ$	TBC
REQ_07	≤ 10 atm	Calibrated pressure gauge	$P \leq 10.13$ bar	PASS
REQ_08	Rectus Type 26 coupling	Physical coupling test	Mates without force; seals at 10 bar	PASS
REQ_09	Air + ambient water only	Inspection of inflation source	Correct gas/liquid confirmed	PASS
REQ_10	Remote launch ≥ 5 m	Umbilical length measurement	$L_{\text{umbilical}} \geq 5000$ mm	TBC
REQ_11	Baffle ≤ 50 % blockage	CAD cross-section + CFD	Transverse 48.9 %	PASS
REQ_12	Altimeter housing	Visual + vent check	3×3 mm ² vents, no adhesive, water-sealed	PASS
REQ_13	Budget $\leq \text{€}500$	BOM total incl. buffer	BOM total $\leq \text{€}500$	PASS
REQ_14	Tank presented before integration	Pre-launch checklist	Judges confirm volume	TBC
REQ_15	Maximise F_S	Flight data	$F_S > 15$ (target: 18.41)	TBC
REQ_16	≥ 3 reusable launches	Post-flight structural check	No critical damage after each flight	TBC
REQ_17	Min. cross-section ≥ 50 %	CAD measurement	Open area = 51.1 %	PASS

5.4 Design Compliance Matrix

Table 25 presents the final Design Compliance Matrix, summarising every mandatory requirement with the achieved specification and a clear PASS/FAIL status.

Table 25: Design Compliance Matrix.

ID	Requirement	Achieved Spec.	Explanation of Design Choice	Status
REQ_01	Total length ≤ 1.5 m	951 mm	Tandem configuration kept compact to minimise bending moments.	PASS
REQ_02	Wingspan ≤ 2.25 m	556 mm	Sized for optimal L/D while easily fitting launch rail clearances.	PASS
REQ_03	Takeoff mass ≤ 5 kg	≈ 2.0 kg	Lightweight PET/PLA materials used to maximise TWR.	PASS
REQ_04	Sloshing tank ≥ 500 mL at 50 %	750 mL at 50 %	Dedicated 1.5 L PET bottle selected for maximum volume allowance.	PASS
REQ_05	Propellant $\leq$ sloshing mass	750 g	Matched exactly 1:1 to maximise thrust without violating rules.	PASS
REQ_06	Launch angle $85 \pm 3^\circ$	Configurable	Rail angle set by ground support equipment on launch day.	TBC
REQ_07	Pressurisation ≤ 10 atm	10 bar (9.87 atm)	Maximum allowed pressure targeted to maximise ascent altitude.	PASS
REQ_08	Rectus Type 26 coupler	Installed	Custom PLA threaded adapter bonded with marine epoxy for security.	PASS
REQ_09	Air + ambient water only	Confirmed	No chemical additives; strictly compliant with competition safety.	PASS
REQ_10	Remote launch ≥ 5 m	Umbilical ≥ 5 m	Standard RC receiver / servo release mechanism ensures safe distance.	TBC
REQ_11	Baffle blockage ≤ 50 %	48.9 % transverse	Geometry explicitly optimised (inner $R = 35$ mm) to stay under limit.	PASS
REQ_12	Altimeter housing limits	O-ring, 3x3 mm vents	Sealed mechanically to avoid disallowed adhesives on the sensor body.	PASS
REQ_13	Budget $\leq \text{€}500$	435.60 €	COTS hardware and 3D printing utilised to keep costs well below limit.	PASS
REQ_14	Tanks presented empty	Checklist item	Pre-flight procedures mandate judge verification prior to integration.	TBC
REQ_15	Maximise flight score	$F_S = 18.41$	Configuration C (deployable wings) chosen strictly to maximise this score.	TBC
REQ_16	≥ 3 reusable launches	Conservative design	Marine epoxy and high-infill PLA chosen for high impact durability.	TBC
REQ_17	Min cross-sec ≥ 50 % of max	51.1 % open area	Symmetrical cruciform/annular gaps guarantee fluid bypass.	PASS

6 Conclusion

This Final Design Report presents the comprehensive engineering development of Project Buzz Lightyear, a tandem delta-wing water rocket designed to maximise the ASRW 2026 Flight Score. Building on the 85/100 Conceptual Design Report, the following outcomes were achieved:

- **Configuration Selection:** Configuration C (deployable main wings and secondary surfaces) was confirmed as the superior design architecture. By stowing the lifting surfaces during the boost phase

to minimize drag and deploying them for descent, the vehicle achieves a predicted flight score of 18.41. The extended glide phase significantly increases horizontal range to 18.28 m, satisfying the primary competition objective.

- **Sloshing Mitigation:** The unbaffled CFD simulation established a baseline lateral sloshing frequency of 3.15 Hz, demonstrating excellent agreement with the 3.11 Hz analytical prediction. The final **Mixed baffle configuration** (annular rings integrated with a central PLA cruciform mast) reduces peak lateral sloshing forces by **72.7 %** (from 15.4 N to 4.2 N) relative to the unbaffled baseline. This successfully exceeds the team's success criterion of a $> 50\%$ force reduction. The Mixed design proved significantly superior to a purely Annular configuration by effectively partitioning the convective liquid mass and suppressing the dominant lateral modes without violating REQ-11 blockage limits.
- **Structural and Pressure Integrity:** The propulsion system plumbing is analytically verified to withstand 10 bar operating pressure with a safety factor > 2 . The Rectus Type 26 coupler provides a safety margin of $> 5\times$. The sloshing tank assembly was streamlined by cutting and epoxy-sealing the base, eliminating complex internal snap-fit joints and ensuring a rigid, leak-proof baffle installation.
- **Requirements Compliance:** 12 of 17 requirements are fully verified at FDR stage (PASS); 5 remain pending physical testing at the Finals.

Lessons Learned. The most significant challenge encountered was the internal baffling of the restrictive 28 mm bottle neck. This constraint forced a pivot from internal interlocking parts to a much more robust base-cutting and marine-epoxy sealing method. Future teams should consider designing modular tanks where base access is planned from the conceptual stage. Additionally, the initial analytical sloshing frequency formulas assume perfect cylinders; mapping these early to the actual tapered PET geometry in CFD prevents downstream discrepancy errors.

Future Work. Pending activities before the Finals include: physical hydrostatic pressure test of the propulsion assembly; wing bond strength test; functional altimeter housing water immersion test; and at least one full launch demonstration if facilities permit.

Final Data Sheet

The following data sheet summarizes the finalized key parameters of the vehicle, establishing the baseline for mass properties, overall dimensions, stability characteristics, and predicted aerodynamic performance.

Table 26: Final vehicle data sheet summary.

Parameter	Value
Mass Properties	
Total Wet Mass (takeoff)	2.0 kg
Dry Mass	1.25 kg
Final Dimensions	
Total Length	951 mm
Body Diameter	94.4 mm
Full Wingspan	556 mm
Aerodynamics & Stability	
Center of Gravity (CG) at launch	508 mm from nose
Center of Pressure (CP)	681 mm from nose
Predicted Performance	
Predicted Apogee	38.94 m
Top Speed	27 m/s

References

- [1] Abramson, H. N. (1966). *The Dynamic Behavior of Liquids in Moving Containers*. NASA SP-106. National Aeronautics and Space Administration.
- [2] Burgarella, F. (2026). ASRW 2026 Sessions 4–5: Sloshing Simulations with Ansys Fluent. *ANSYS / Synopsys CFD Applications Engineering Webinar*, March 2026.
- [3] Water Rocket Simulator (WRS). Retrieved from <https://waterrocketsimulator.github.io>
- [4] Manfredo, J. (2005, October 7). Are your descent rates decent?: Calculating how fast your models fall. *Apogee Components Peak of Flight*, 149, 1–4. Retrieved from www.ApogeeRockets.com
- [5] Mark, J. E. (Ed.). (2007). *Physical Properties of Polymers Handbook* (2nd ed.). Springer.
- [6] Dodge, F. T. (2000). *The New Dynamic Behavior of Liquids in Moving Containers*. Southwest Research Institute.
- [7] Patkas, L. A., & Karamanos, S. A. (2007). *Variational Solutions for Externally Induced Sloshing in Horizontal-Cylindrical and Spherical Vessels*. *Journal of Engineering Mechanics*, 133, 641–655.
- [8] Sun, Y., Zhou, D., Wang, J., Han, H. (2021). *Lumped Parameter Model for Liquid Sloshing in a Cylindrical Tank Equipped with Multiple Annular Baffles*. *Journal of Structural Engineering*, 147.
- [9] ANSYS, Inc. (2025). *ANSYS Meshing User's Guide*, Release 2025 R1, “Mesh Quality / Mesh Metrics” section. Canonsburg, PA: ANSYS, Inc. Available at: https://ansyshelp.ansys.com/public/Views/Secured/corp/v251/en/wb_msh/msh_check_mesh_quality.html (accessed 13 June 2026).
- [10] Fleeman, E. L. (2006). *Tactical Missile Design*, 2nd ed. AIAA Education Series. Reston, VA: American Institute of Aeronautics and Astronautics. ISBN 978-1-56347-782-9.
- [11] *Integrated Missile Fin Deployment System*, U.S. Patent No. 6,073,880, U.S. Patent and Trademark Office, 2000. Available at: <https://patents.google.com/patent/US6073880A>.
- [12] Roache, P. J. (1994). Perspective: A Method for Uniform Reporting of Grid Refinement Studies. *Journal of Fluids Engineering*, 116(3), 405–413. <https://doi.org/10.1115/1.2910291>
- [13] Schlichting, H., & Gersten, K. (2000). *Boundary-Layer Theory* (8th ed.). Springer, Berlin, Heidelberg. <https://doi.org/10.1007/978-3-642-85829-1>
- [14] Hoerner, S. F. (1965). *Fluid-Dynamic Drag: Practical Information on Aerodynamic Drag and Hydrodynamic Resistance*. Hoerner Fluid Dynamics, Midland Park, NJ. Available at: <https://archive.org/details/FluidDynamicDragHoerner1965>

A Full Risk Assessment Matrix

The complete ASRW 2026 Risk Assessment, completed using the provided template, is included below. Risks are scored on a 5×5 likelihood–severity matrix. Risks rated ≥ 15 (High) require additional mitigation prior to the Finals.

B Trajectory Simulation Input Parameters

Table 27: Full input parameter table for the water rocket trajectory simulation.

Parameter	Value	Unit	Config.
<i>Environmental Constants</i>			
Gravity g	9.81	m/s^2	A, B, C
Air density ρ_a	1.225	kg/m^3	A, B, C
Water density ρ_w	1000	kg/m^3	A, B, C
<i>Rocket Geometry & Mass</i>			
Vessel capacity V	2.25	L	A, B, C
Dry mass m_r	1.25	kg	A, B, C
Propellant water mass m_w	0.75	kg	A, B, C
Body diameter d	94.4	mm	A, B, C
Nozzle diameter d_e	28.0	mm	A, B, C
Launch angle θ	85	$^\circ$	A, B, C
<i>Aerodynamic Coefficients</i>			
$C_{D0,A}$	0.010	—	A, C
$C_{D0,p}$ (parachute)	0.750	—	A
$C_{D0,e}$ (equivalent)	0.748	—	A (descent)
$C_{D0,B}$	0.016	—	B, C
<i>Areas</i>			
Rocket section A	0.007	m^2	A, B
Parachute area A_p	0.554	m^2	A
Wing area A_{wing}	0.16	m^2	B, C (glide)
<i>Propulsion</i>			
Operating pressure P_0	10.0	bar	A, B, C
Burnout time t_b	0.035	s	A, B, C
Air phase end t_a	0.125	s	A, B, C

C Sloshing CFD Named Expressions

The following Named Expressions were entered into ANSYS Fluent to implement the time-varying gravity body-force profile:

```
; --- Effective acceleration magnitude ---
a_eff = IF(Flow_Time <= 0.125 [s], 757.33 [m/s^2], 9.81 [m/s^2])

; --- Tank attitude angle (degrees, capped at -79.79°) ---
theta_deg = MAX(85 [deg] - 28.11 [deg/s] * Flow_Time, -79.79 [deg])

; --- Body force components ---
gravity_x = -a_eff * sin(theta_deg)
gravity_y = 0 [m/s^2]
gravity_z = -a_eff * cos(theta_deg)
```

These expressions are assigned to the *Gravity* body-force specification in the Fluent solver under *General* → *Gravity*.